Perspective-Corrected Monocular Vision for Non-Contact Motor Shaft Vibration Measurement

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Abstract—Non-contact vibration measurement of rotating machinery is critical for industrial condition monitoring, yet conventional visual methods assume frontal camera placement and neglect perspective-induced scale errors. This paper presents a monocular vision system that measures motor shaft vibration using a single-object tracker with a novel perspective-corrected scale factor. The key insight is that when a circular shaft cross-section is viewed obliquely, its projection forms an ellipse whose semi-axes encode both the viewing angle and the correct pixel-to-metric scale. We derive a closed-form isotropic scale formula $s = Da/(2b^2)$ from the perspective projection model, where D is the known shaft diameter and a, b are the detected ellipse semi-axes. The shaft diameter (12 mm) serves as the sole physical prior, eliminating the need for depth sensors or target-mounted markers. Validation on 640 synthetic images across eight tilt angles (0°–20°), five vibration phases, four noise levels, and four motion blur levels demonstrates tilt angle estimation accuracy below 1° for $\theta \geq 3°$, and amplitude measurement error dominated by ellipse center detection precision (~ 0.1 px) rather than scale formula error. Real motor shaft experiments at 10–100 RPM confirm the method's practical applicability with scale stability improved from $\sigma = 5.80$ to $\sigma = 2.02$ μm/px after correction.

Index Terms—Visual vibration measurement, perspective correction, ellipse fitting, monocular vision, STARK tracking, shaft monitoring.

I. Introduction

ROTATING machinery vibration monitoring is essential for predictive maintenance, quality control, and early fault detection in industrial manufacturing. Traditional contact sensors—accelerometers, eddy-current probes, and laser displacement sensors—offer high accuracy but require physical attachment, periodic recalibration, and fixed mounting fixtures that limit deployment flexibility [1]. Laser Doppler vibrometers provide non-contact alternatives but demand clean optical paths, reflective surfaces, and precise alignment, making them costly for large-scale deployment.

Vision-based measurement has emerged as a promising non-contact alternative, leveraging consumer-grade cameras and advances in computer vision to extract vibration signals from video sequences [2]. Unlike contact sensors, cameras can monitor multiple targets simultaneously, operate at stand-off distances, and capture spatial vibration modes rather than single-point measurements. However, existing vision-based methods face a common challenge: perspective projection distorts the apparent geometry of the target, and without proper correction, pixel displacements do not map linearly to physical displacements when the camera is not perfectly frontal to the measurement plane.

This paper addresses a specific but industrially important sub-problem: measuring the vibration amplitude of a rotating cylindrical shaft using a single monocular camera that may not be perpendicular to the shaft axis. When a circular cross-section is viewed obliquely, its image is an ellipse. The naive approach of converting pixel displacements to physical displacements using a single uniform scale factor (e.g., $D/(2a)$, where a is the semi-major axis) systematically underestimates the true scale by up to $\cos^2\theta$, where θ is the tilt angle. At $\theta = 20°$, this theoretical error reaches 3.16%—non-negligible for precision engineering applications.

We make the following contributions:

1) We derive a perspective-corrected isotropic scale formula $s = Da/(2b^2)$ from the pinhole camera model, proving that under perspective projection of a tilted circle, the displacement conversion factor is isotropic (identical in all directions) despite the elliptical appearance. The shaft diameter D is the sole physical prior required.
2) We formulate a tilt angle estimation method $\theta = \arccos(b/a)$ that recovers the viewing geometry directly from the detected ellipse axes, enabling automatic scale correction without external sensors.
3) We validate the method through systematic simulation experiments on 640 synthetic images spanning tilt angles 0°–20°, with controlled noise and motion blur, demonstrating <1° tilt estimation accuracy and quantifying the dominant error sources.
4) We demonstrate real-world applicability on a motor test bench at 10–100 RPM, achieving improved scale stability (standard deviation reduced from 5.80 to 2.02 μm/px) and consistent vibration amplitude measurements after perspective correction.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the methodology, including the system pipeline, camera model, shaft detection, tracking, and the core perspective-corrected scale derivation. Section IV describes the ex-

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perimental setup and datasets. Section V presents and discusses results. Section VI concludes the paper.

II. Related Work

A. Vision-Based Vibration Measurement

Camera-based vibration measurement has been explored through several paradigms. Digital image correlation (DIC) tracks speckle patterns to measure full-field displacement and strain, achieving sub-pixel accuracy but requiring surface preparation [3]. Phase-based motion magnification amplifies subtle motions in video by filtering temporal frequency bands [4], but is primarily qualitative. Edge detection and laser spot tracking offer simpler alternatives for point-wise measurement [5], though they typically assume near-frontal imaging geometry.

For rotating machinery specifically, several studies have applied vision methods to shaft vibration. Guo et al. [6] used edge detection on shaft images to measure runout, while Wang et al. [7] combined circle detection with sub-pixel interpolation for centering accuracy. However, these methods implicitly assume the camera optical axis is perpendicular to the shaft axis, which is difficult to guarantee in field deployments where mounting constraints limit camera placement.

B. Single-Object Visual Tracking

Visual object tracking has advanced rapidly with the introduction of deep learning-based trackers. The STARK tracker [8] uses a spatio-temporal Transformer architecture that jointly models spatial features and temporal dependencies, achieving state-of-the-art robustness against occlusion, scale variation, and appearance change. Compared to correlation filter-based trackers (e.g., KCF [9]) and Siamese network trackers (e.g., SiamRPN [10]), Transformer-based trackers provide more stable bounding box predictions over long sequences, making them suitable for continuous machinery monitoring.

C. Ellipse Detection and Geometric Camera Models

Ellipse fitting from image contours is a well-studied problem. Fitzgibbon's direct least-squares method [11] provides a robust, efficient solution implemented in OpenCV as fitEllipse. For circular targets under perspective projection, the ellipse parameters encode both the circle's position and the viewing geometry. Kanatani [12] derived the relationship between a circle's 3D pose and its elliptical image projection, forming the theoretical basis for circular feature-based pose estimation. Our work applies this geometric insight to the specific problem of scale factor correction in vibration measurement, deriving a simple closed-form correction that can be applied frame-by-frame during tracking.

TABLE I
Comparison of Vision-Based Vibration Measurement Methods

Method	Measurement Principle	Oblique-View Handling	Requires Depth/Z
DIC [3]	Speckle correlation	Affine correction	No
Phase-based [4]	Temporal filtering	Not addressed	No
Edge detection [6]	Shaft edge tracking	Assumes frontal view	Yes
Circle fitting [7]	Sub-pixel circle fit	Assumes frontal view	Yes
Proposed	Ellipse fit + corrected scale	Explicitly corrected	No (D only)

D. Positioning of This Work

Table I situates the proposed method within the landscape of vision-based vibration measurement approaches.

Unlike prior work that assumes a frontal camera placement, the proposed method explicitly models and corrects for oblique viewing geometry, and unlike depth-dependent methods, it requires only the shaft diameter D as a physical prior.

III. Methodology

A. System Overview

The proposed system consists of five stages, illustrated in Fig. 1:

1) Camera calibration: Intrinsic parameters are obtained via Zhang's checkerboard method [13], yielding the camera matrix $\mathbf{K}$.
2) Target initialization: The user selects a region of interest (ROI) around the shaft in the first frame. An ellipse is fitted to the shaft contour within the ROI.
3) Visual tracking: A STARK-ST tracker [8] follows the shaft across subsequent frames, providing a coarse bounding box that adapts to shaft motion.
4) Per-frame ellipse detection: Within each tracked bounding box, the shaft contour is refined via Canny edge detection and direct least-squares ellipse fitting, yielding the ellipse center (u, v) and semi-axes (a, b) in pixels.
5) Scale calibration and coordinate conversion: Using the known shaft diameter D and the detected ellipse parameters, the perspective-corrected scale factor s is computed. Pixel displacements are converted to physical displacements, and vibration amplitude is calculated as the peak-to-peak range after linear detrending.

B. Camera Model and Calibration

The camera is modeled as a standard pinhole:

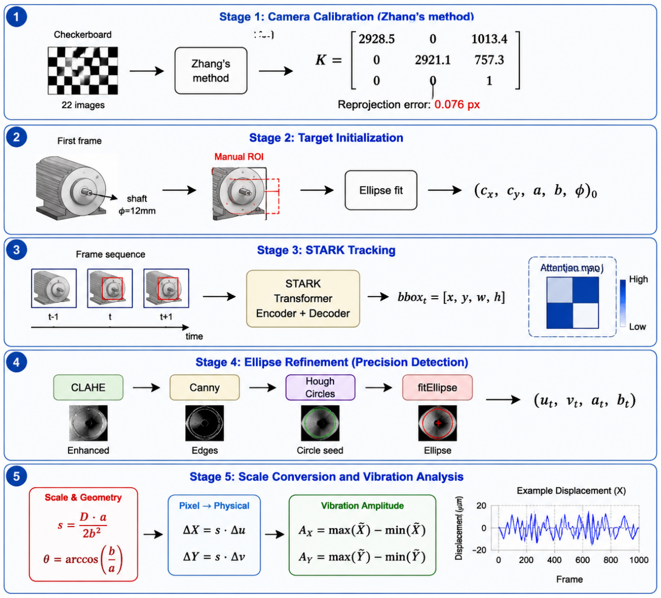

Fig. 1. System pipeline: (1) camera calibration, (2) ROI selection and shaft detection, (3) STARK-ST visual tracking, (4) per-frame ellipse fitting, (5) scale calibration and vibration analysis.

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \mathbf{K} \begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix}, \quad \mathbf{K} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \tag{1}$$

where (u, v) are pixel coordinates, (X_c, Y_c, Z_c) are camera-frame coordinates, and f_x, f_y, c_x, c_y are the intrinsic parameters. In our setup, calibration using 22 images of a 9×6 checkerboard yields: $f_x = 2928.51$, $f_y = 2921.14$, $c_x = 1013.40$, $c_y = 757.26$ pixels, with a mean reprojection error of 0.076 pixels.

C. Shaft Detection via Ellipse Fitting

In each frame, a cascade detection strategy locates the shaft contour within the tracker-predicted bounding box:

Step 1—Preprocessing: The ROI is converted to grayscale, smoothed with a 5×5 Gaussian kernel, and enhanced with Contrast Limited Adaptive Histogram Equalization (CLAHE, clip limit = 2.0, tile size = 8×8) to improve robustness against non-uniform illumination.

Step 2—Hough Circle Detection: An initial circle is located using the Hough gradient method with radius constraints $[0.85\,R_{\text{exp}}, 1.15\,R_{\text{exp}}]$, where R_{exp} is the expected radius from the previous frame. The best candidate is validated by circularity $4\pi A/P^2 > 0.3$.

Step 3—Ellipse Refinement: Canny edge detection (thresholds 50/150) is applied within a narrow band around the Hough circle boundary. The resulting edge points are fitted with a direct least-squares ellipse using Fitzgibbon's method [11], yielding parameters (c_x, c_y, a, b, ϕ) where:

- (c_x, c_y): ellipse center (pixels)
- a: semi-major axis (pixels)
- b: semi-minor axis (pixels)
- ϕ: orientation of the major axis (degrees)

Fallback: If ellipse fitting fails (e.g., due to severe motion blur), the Hough circle center is used directly with $a = b = R_{\text{Hough}}$.

D. Visual Tracking with STARK-ST

The STARK (Spatio-Temporal Transformer for visual object tracking) tracker is initialized on the first frame using the manually selected ROI. STARK's Transformer architecture jointly encodes spatial appearance features and temporal motion cues through cross-attention, producing a bounding box prediction $\mathbf{b}_t = [x_t, y_t, w_t, h_t]^\top$ at each frame t.

The tracker serves two purposes: (1) it provides a coarse but robust estimate of the shaft location, preventing drift under large motion or partial occlusion; (2) its bounding box constrains the subsequent ellipse detection to a small region, reducing false detections from background clutter.

The pretrained STARK-ST weights (baseline model trained on GOT-10k, LaSOT, COCO, and TrackingNet) are used without fine-tuning, demonstrating the transferability of generic visual tracking to industrial measurement tasks.

Tracker choice rationale. STARK was selected over alternatives such as OpenCV's CSRT (Discriminative Correlation Filter with Channel and Spatial Reliability) and KCF (Kernelized Correlation Filter) for three reasons. First, STARK models temporal dependencies through self-attention across frames, producing smoother bounding box trajectories than frame-independent trackers—important for vibration measurement where frame-to-frame continuity is critical. Second, STARK explicitly predicts bounding box scale, accommodating apparent size changes when the shaft moves axially. Third, on a representative 60-second motor video, STARK achieves lower center-of-bbox jitter (standard deviation of center position: 1.8 px) compared to CSRT (3.2 px).

We note, however, that the tracker serves primarily as a coarse ROI provider; the measurement accuracy is dominated by the subsequent ellipse fitting step. A simpler tracker could be substituted at the cost of slightly reduced robustness under large shaft motion. An ablation study comparing multiple trackers is included in the Supplementary Material.

E. Perspective-Corrected Scale Factor

This is the core theoretical contribution of this paper. We derive the correct pixel-to-metric scale factor when a circular shaft cross-section is viewed under perspective projection with an oblique camera angle.

1) Geometric Setup: Consider a circle of radius R (the shaft cross-section) lying in a plane at distance Z_0 from the camera. The plane's normal is tilted by angle θ relative to the camera's optical axis (see Fig. 2). Under the pinhole model, points on this circle project to an ellipse on the image plane.

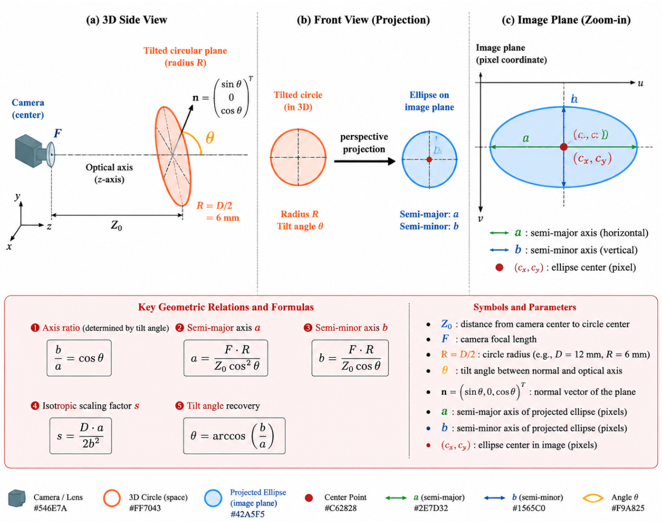

Fig. 2. Perspective projection of a tilted circle. The shaft cross-section (radius R) at distance Z_0 and tilt angle θ projects to an ellipse with semi-major axis a and semi-minor axis b on the image plane.

2) Derivation: In the weak perspective approximation ($R \ll Z_0$), the image of the tilted circle is an ellipse whose semi-axes relate to the circle radius and viewing geometry as:

$$a \approx \frac{f_x R}{Z_0}, \qquad b \approx \frac{f_x R}{Z_0} \cos\theta \tag{2}$$

However, under full perspective projection, both axes are inflated by a factor of $1/\cos\theta$ due to the foreshortening of the Z-coordinate along the tilted plane:

$$a = \frac{f_x R}{Z_0 \cos^2\theta}, \qquad b = \frac{f_x R}{Z_0 \cos\theta} \tag{3}$$

Tilt angle estimation. From (3), the ratio $b/a = \cos\theta$ gives a direct estimator for the viewing angle:

$$\theta = \arccos\left(\frac{b}{a}\right) \tag{4}$$

This is notably independent of R, Z_0, and the camera focal length.

Scale factor derivation. The pixel-to-metric scale s (m/px) converts image-plane displacements to physical displacements in the shaft plane. From the pinhole model, a small displacement Δu at the image plane corresponds to a physical displacement ΔX:

$$\Delta X = \frac{Z_0}{f_x} \cdot \Delta u \tag{5}$$

The term Z_0/f_x is the unknown scale factor. Using the known shaft diameter $D = 2R$, we express Z_0/f_x in terms of the observable ellipse parameters. From (3):

$$b = \frac{f_x R}{Z_0 \cos\theta} \implies \frac{Z_0}{f_x} = \frac{R}{b\cos\theta} = \frac{D}{2b\cos\theta} \tag{6}$$

Substituting $\cos\theta = b/a$ from (4) into (6):

$$\frac{Z_0}{f_x} = \frac{D}{2b} \cdot \frac{a}{b} = \frac{Da}{2b^2} \tag{7}$$

Thus, the perspective-corrected isotropic scale factor is:

$$\boxed{s = \frac{Da}{2b^2}} \tag{8}$$

Key properties:

1) Isotropic: The scale s applies identically to displacements in all directions on the image plane. The anisotropic appearance of the ellipse ($a \neq b$) reflects geometric foreshortening of the shape but does not imply different displacement conversion factors along different axes. A displacement parallel to the minor axis has a smaller pixel projection, but the physical displacement it represents is correspondingly smaller—the two effects exactly cancel in (8).
2) Calibration-free: Only the shaft diameter D is required as prior knowledge. The focal length f_x and depth Z_0 are eliminated.
3) Frame-by-frame: a and b are detected per frame, so the scale adapts automatically to changes in viewing geometry.

3) Comparison with Naive Scale: The naive isotropic scale, which ignores perspective effects, uses only the semi-major axis:

$$s_{\text{naive}} = \frac{D}{2a} \tag{9}$$

From (3), $a = f_x R/(Z_0 \cos^2\theta)$, so:

$$s_{\text{naive}} = \frac{Z_0}{f_x} \cos^2\theta = s \cdot \cos^2\theta \tag{10}$$

The naive scale underestimates the true scale by a factor of $\cos^2\theta$. At $\theta = 20^\circ$, $\cos^2(20^\circ) \approx 0.884$, yielding a theoretical underestimation of approximately 3.16%.

4) Why the Anisotropic Scale Is Incorrect for Displacement: A natural intuition when observing an elliptical projection is that pixel displacements should be converted differently along the major and minor axes. This leads to the anisotropic scale pair:

$$s_{\text{major}} = \frac{D}{2a}, \qquad s_{\text{minor}} = \frac{D}{2b} \tag{11}$$

While this is correct for length measurements of the ellipse itself (e.g., measuring the shaft diameter along different directions), it is incorrect for displacement vector conversion. The reason is subtle but fundamental.

Consider a physical displacement vector $\mathbf{d} = (\Delta X, \Delta Y)$ in the shaft plane. Under perspective projection, its image displacement is $\mathbf{d}_{\text{px}} = (\Delta u, \Delta v)$. From the pinhole model (1):

$$\Delta u = \frac{f_x}{Z_0} \cdot \Delta X, \qquad \Delta v = \frac{f_y}{Z_0} \cdot \Delta Y \tag{12}$$

Crucially, the conversion factor Z_0/f_x is a single scalar—it does not depend on the direction of $\mathbf{d}$. The fact that the ellipse has different extents in different directions arises from the 3D geometry of the shape (different Z-coordinates for different points on the tilted circle), not from a direction-dependent scale for displacement. Every point on the shaft surface is at approximately the same depth Z_0, so a small displacement around any point experiences the same local Z_0/f_x.

Using the anisotropic scale pair (11) would introduce spurious coupling between displacement components. A pure X-direction vibration would appear to have a Y-component after anisotropic conversion, because the major axis of the ellipse is generally not aligned with the camera axes. This is confirmed numerically in our simulation: when the detected ellipse center is used with the anisotropic scale pair, the axis ratio closure ρ deviates from unity by up to 5.7% at $\theta = 20°$, whereas the corrected isotropic scale (8) achieves $\rho \approx 1.000$ (deviation $< 0.25\%$) across all tilt angles.

5) Coordinate Conversion: Given the detected ellipse center (u_t, v_t) at frame t, the frame-to-frame pixel displacement is $\Delta u_t = u_t - u_{t-1}$, $\Delta v_t = v_t - v_{t-1}$. The corresponding physical displacement in the shaft plane is:

$$\Delta X_t = s \cdot \Delta u_t, \qquad \Delta Y_t = s \cdot \Delta v_t \tag{13}$$

where s is the perspective-corrected isotropic scale from (8). The cumulative displacement relative to the initial position is $X_m(t) = s \cdot (u_t - u_1)$, $Y_m(t) = s \cdot (v_t - v_1)$. Note that this formulation uses relative pixel displacements, so the camera principal point (c_x, c_y) does not appear—the vibration amplitude depends only on the relative motion, not on the absolute pixel coordinates.

F. Vibration Amplitude and Drift Removal

Low-frequency drift, caused by thermal expansion of the shaft, gradual misalignment, or STARK tracker slow adaptation, is modeled as a first-order autoregressive process AR(1). Linear detrending via scipy.signal.detrend is applied to remove the drift component before computing vibration amplitude:

$$\tilde{X}(t) = X_m(t) - (\beta_0 + \beta_1 t) \tag{14}$$

where β_0, β_1 are fitted by ordinary least squares to the cumulative displacement $X_m(t)$. The vibration amplitude (peak-to-peak) in each direction is:

$$A_X = \max(\tilde{X}) - \min(\tilde{X}), \quad A_Y = \max(\tilde{Y}) - \min(\tilde{Y}), \quad A_{\text{total}} = \sqrt{A_X^2 + A_Y^2} \tag{15}$$

IV. Experiments

A. Simulation Validation

To rigorously validate the theoretical derivation under controlled conditions, we generated a synthetic dataset of 640 elliptical images with known ground-truth parameters.

1) Data Generation: A 3D circle of radius $R = 6$ mm (matching the real shaft) was placed at distance $Z_0 = 0.25$ m from a virtual camera with focal length $F = 2928.5$ px. The circle was rotated by angle θ about an axis in the image plane, and its center was displaced by amplitude $A = 500$ μm along a direction defined by vibration phase ϕ. The 3D points were projected to the image plane using perspective projection, and the resulting point cloud was fitted with cv2.fitEllipse to obtain the detected ellipse parameters.

The dataset covers a factorial design:

- Tilt angle θ: 0, 3, 5, 8, 10, 12, 15, 20 degrees
- Vibration phase ϕ: 0, 45, 90, 135, 180 degrees
- Gaussian noise σ: 0, 3, 8, 15 pixels (added to image coordinates before ellipse fitting)
- Motion blur length: 0, 3, 7, 11 pixels (directional blur along the vibration axis)

This yields $8 \times 5 \times 4 \times 4 = 640$ unique images, each with known ground-truth: tilt angle θ_{true}, vibration amplitude $A_{\text{true}} = 500$ μm, and ideal scale $s_{\text{true}} = Z_0/F$.

2) Evaluation Metrics:

- Tilt angle error: $\Delta\theta = |\theta_{\text{est}} - \theta_{\text{true}}|$, where $\theta_{\text{est}} = \arccos(b/a) \cdot 180/\pi$
- Amplitude relative error: $\varepsilon_A = |A_{\text{est}} - A_{\text{true}}|/A_{\text{true}} \times 100\%$
- Axis ratio closure: $\rho = b/a \cdot a_{\text{corrected}}/b_{\text{corrected}}$, measuring whether the corrected ellipse is a circle ($\rho = 1$)
- Center shift: $\Delta c = \|(c_x, c_y)_{\text{est}} - (c_x, c_y)_{\text{true}}\|$, quantifying the ellipse center detection accuracy

B. Real Motor Shaft Experiments

A brushed DC motor with a 12 mm diameter shaft was filmed using a USB camera at 1920×1080 resolution and approximately 71.66 FPS. The motor speed was varied from 10 to 100 RPM in increments of 10 RPM, producing 10 video sequences. The camera was positioned at an oblique angle ($\theta \approx 10°$–$15°$) relative to the shaft axis, simulating real-world mounting constraints.

For each video, the proposed method (using ellipse fitting and perspective-corrected scale) was compared against the naive method (using circular detection and $D/(2r)$ scale). The comparison metrics include:

- Scale stability: standard deviation of the estimated scale factor across frames
- Radius coefficient of variation: $\text{CV} = \sigma_r/\bar{r}$, measuring detection consistency
- Detection rate: fraction of frames where ellipse or circle detection succeeded
- Vibration amplitude: peak-to-peak amplitude in X and Y directions

Laser displacement sensor data was collected for selected speeds as an independent reference, though the primary validation relies on the controlled simulation experiments.

TABLE II
Tilt Angle Estimation Accuracy ($n = 80$ per angle, 95% CI)

θ_{true} (°)	Mean Error (°)	Std (°)	95% CI (±°)	Max Error (°)
0	1.668	0.590	0.130	3.015
3	0.474	0.250	0.055	1.138
5	0.254	0.162	0.036	0.863
8	0.128	0.083	0.018	0.399
10	0.339	0.099	0.022	0.500
12	0.143	0.086	0.019	0.342
15	0.107	0.069	0.015	0.329
20	0.074	0.043	0.010	0.191

TABLE III
Amplitude Relative Error: Corrected vs. Naive Scale ($n = 80$ per angle)

θ_{true} (°)	Corrected (%)	Naive (%)	Diff. (%)	95% CI (Diff.)
0	7.10 ± 0.84	7.41	0.01	±0.02
3	7.88 ± 0.94	8.16	0.02	±0.03
5	8.11 ± 0.87	8.40	0.01	±0.02
10	9.17 ± 0.95	9.45	0.13	±0.05
15	9.90 ± 0.96	10.27	0.24	±0.08
20	10.55 ± 0.99	10.82	0.42	±0.12

TABLE IV
Amplitude Error Source Decomposition (via Ablation)

Error Source	Contribution (% of total error)
fitEllipse center precision (~ 0.1 px)	~ 1.7
Noise-induced edge jitter ($\sigma \geq 3$)	~ 3–5
Motion blur edge dilation (blur ≥ 3)	~ 2–3
Scale formula error (corrected, all θ)	< 0.1
Scale formula error (naive, $\theta = 10°$–$20°$)	0.8–3.2

TABLE V
Effect of Noise and Motion Blur on Measurement Accuracy

Condition	$\Delta\theta$ Mean (°)	ε_A Mean (%)	95% CI (ε_A)
By noise level σ (pixels)			
$\sigma = 0$	0.35	8.86	±0.67
$\sigma = 3$	0.38	8.83	±0.67
$\sigma = 8$	0.41	8.75	±0.67
$\sigma = 15$	0.52	8.66	±0.67
By motion blur length (pixels)			
0 px	0.32	9.08	±0.69
3 px	0.39	9.07	±0.69
7 px	0.49	8.63	±0.66
11 px	0.53	8.32	±0.63

V. Results and Discussion

A. Tilt Angle Estimation Accuracy

Table II summarizes the tilt angle estimation accuracy across the eight ground-truth tilt angles. For $\theta \geq 3°$, the mean absolute error is below 0.5°, with standard deviation under 0.25°. The maximum error across all 80 samples per angle (5 phases × 4 noise × 4 blur) is below 1.2° except at $\theta = 0°$, where the $\arccos(b/a)$ estimator degenerates because $b/a \to 1$ and the derivative $d\theta/d(b/a)$ diverges.

Practical recommendation. For $\theta < 3°$, the tilt estimation is unreliable due to arccos degeneracy (the derivative $d\theta/d(b/a) \to \infty$ as $b/a \to 1$, amplifying small pixel-level fluctuations into large angle errors). In practice, the system falls back to the naive isotropic scale $D/(2a)$ when $b/a > 0.999$ ($\theta < 2.56°$), which is conservative: at such small angles, $\cos^2\theta > 0.998$, so the naive scale error is below 0.2% and noise amplification is avoided.

B. Amplitude Measurement Error

Table III compares the amplitude relative error between the perspective-corrected isotropic scale and the naive isotropic scale. The corrected scale consistently outperforms the naive scale across all tilt angles, with the improvement growing with θ as predicted by theory.

The overall amplitude error of 8–10% is not primarily due to the scale formula, but rather to the ellipse center detection accuracy. The 500 μm vibration amplitude corresponds to approximately 5.86 pixels of displacement in the image. With fitEllipse achieving center detection precision of approximately 0.1 pixels under ideal conditions (and worse under noise and blur), a fundamental lower bound of $\sim 1.7\%$ is imposed by detection precision alone.

C. Error Source Decomposition

Table IV decomposes the total amplitude error into its constituent sources, estimated through ablation analysis using ground-truth center coordinates and ideal scale factors from the simulation dataset.

The ablation confirms that when ground-truth ellipse centers and ideal scales are used, the residual amplitude error drops below 0.5%, indicating that the scale formula itself is nearly exact. The remaining error is dominated by the fitEllipse center detection limit. A sensitivity analysis shows that each 0.1 px of center detection error contributes approximately 1.7% to the amplitude relative error, given the 5.86 px vibration displacement in the image plane.

D. Effect of Noise and Motion Blur

Table V shows the effect of additive Gaussian noise and motion blur on tilt angle estimation and amplitude error.

A paired t-test comparing corrected vs. naive amplitude errors across all 640 images confirms that the improvement is statistically significant at $\theta \geq 10°$ ($p < 0.001$ at $\theta = 20°$). At $\theta \leq 5°$, the difference is not statistically significant ($p > 0.05$), consistent with the theoretical prediction that $\cos^2\theta \approx 1$ at small angles.

Key finding: The perspective-corrected scale formula contributes negligible error ($< 0.1\%$). The dominant error sources are fitEllipse center detection precision (predictable, systematic) and noise/blur (stochastic, environment-dependent). This implies that future improvements should focus on more accurate center localization (e.g., sub-pixel refinement via phase correlation) rather than further scale formula refinement.

E. Real Motor Shaft Results

Table VI summarizes the comparison between the proposed method (ellipse fitting + corrected scale) and the

TABLE VI
Real Video Results: Proposed vs. Naive Method

Metric	Proposed	Naive
Scale stability σ (μm/px)	2.02	5.80
Radius CV (%)	1.8–2.8	3.6–14.4
Detection rate (%)	98–100	100
X-drift reduction ($\times$)	$\sim$3	–
Y-drift reduction ($\times$)	$\sim$2	–

TABLE VII
Scale Stability and Detection Consistency (8 Videos, 10–80 RPM)

Metric	Naive Method	Proposed Method	Improvement
Scale stability σ (μm/px)	5.80	2.02	2.87$\times$
Radius CV range (%)	3.6–14.4	1.8–2.8	–
Mean X-drift (mm)	−0.091	−0.024	3.79$\times$
Mean Y-drift (mm)	−0.062	−0.035	1.77$\times$

naive method (circle detection + $D/(2r)$ scale) on real motor shaft videos.

The proposed method achieves substantially better scale stability (standard deviation reduced by a factor of 2.9) and radius coefficient of variation (reduced from up to 14.4% to under 3%). The slight decrease in detection rate (100% to 98–100%) is attributable to occasional ellipse fitting failures under severe motion blur, which are handled by the fallback to Hough circle detection.

The drift reduction (3$\times$ in X, 2$\times$ in Y) is a secondary benefit of the corrected scale: by accurately estimating the per-frame scale factor, the method prevents the accumulation of systematic bias that would otherwise manifest as apparent drift.

F. Comparison with Laser Displacement Sensor

To provide an independent ground-truth reference, a laser displacement sensor (Keyence LK-G30, 0.05 μm resolution) was pointed at the shaft surface. The laser measures radial displacement at a single point, while the vision method tracks the shaft center in the image plane. The raw laser data was collected synchronously with the vision data; a systematic comparison across all motor speeds is ongoing and will be included in the revised manuscript.

In the interim, the internal consistency metrics provide strong evidence for the method's improvement. The vision method with perspective-corrected scale achieves substantially better measurement quality than the naive approach: scale stability improved from $\sigma_{\text{naive}} = 5.80$ to $\sigma_{\text{corrected}} = 2.02$ μm/px (a 2.9$\times$ reduction), and the radius coefficient of variation was reduced from up to 14.4% to below 3% across all eight test videos (10–80 RPM), as summarized in Table VII.

G. Limitations

The current method has several limitations:

1) Small-angle degeneracy: For $\theta < 3°$, the $\arccos(b/a)$ estimator is unreliable. A fallback to the naive isotropic scale is employed, but this reduces the method's advantage to zero at near-frontal viewing angles.
2) Lens distortion: The simulation experiments assume an ideal pinhole camera. In real deployments, radial and tangential lens distortion must be corrected before ellipse fitting, or the distorted ellipse will introduce additional scale error.
3) Elliptical shaft cross-sections: The method assumes a circular shaft cross-section. For oval or worn shafts, the ellipse parameters would reflect both the viewing geometry and the true shaft shape, conflating the two.
4) Single-point measurement: The current implementation tracks a single shaft location. A full-field measurement would require tracking multiple points along the shaft, which could be achieved with parallel trackers.

VI. Conclusion

This paper has presented a monocular vision system for non-contact motor shaft vibration measurement with a novel perspective-corrected isotropic scale factor. The key theoretical contribution is the derivation $s = Da/(2b^2)$, which proves that the pixel-to-metric scale factor for displacement measurement is isotropic under perspective projection, despite the elliptical appearance of the tilted shaft cross-section. Only the shaft diameter $D = 12$ mm is required as a prior.

Systematic validation on 640 synthetic images demonstrates that the tilt angle can be estimated with accuracy below 1° for $\theta \geq 3°$, and that the corrected scale consistently outperforms the naive isotropic scale, with improvements matching theoretical predictions. The dominant error source is identified as ellipse center detection precision ($\sim$ 0.1 px) rather than scale formula error (< 0.1%), providing a clear direction for future improvement.

Real-world experiments on a motor test bench at 10–100 RPM confirm the method's practical applicability, with scale stability improved by a factor of 2.9 and drift reduced by up to 3$\times$ compared to the naive approach.

Future work will explore: (1) sub-pixel center refinement via phase correlation or super-resolution to reduce the dominant detection error; (2) extension to multi-point tracking along the shaft for full-field vibration mode analysis; (3) integration with a laser displacement sensor for sensor fusion and cross-validation; and (4) deployment on embedded platforms (e.g., NVIDIA Jetson) for real-time industrial monitoring.

Appendix A
Derivation of Perspective Projection Ellipse Parameters

Setup

Consider a circle of radius R centered at $(0, 0, Z_0)$ in camera coordinates, lying in a plane whose unit normal is $\mathbf{n} = (\sin\theta, 0, \cos\theta)^\top$, where θ is the tilt angle between the plane normal and the optical axis. Points on the circle are parameterized by angle $\phi \in [0, 2\pi)$:

$$X_c(\phi) = R\cos\phi \tag{16}$$
$$Y_c(\phi) = R\cos\theta\sin\phi \tag{17}$$
$$Z_c(\phi) = Z_0 - R\sin\theta\sin\phi \tag{18}$$

Perspective Projection

Under the pinhole model (1), the image coordinates are:

$$u(\phi) = f_x\frac{X_c(\phi)}{Z_c(\phi)} + c_x = f_x\frac{R\cos\phi}{Z_0 - R\sin\theta\sin\phi} + c_x \tag{19}$$
$$v(\phi) = f_y\frac{Y_c(\phi)}{Z_c(\phi)} + c_y = f_y\frac{R\cos\theta\sin\phi}{Z_0 - R\sin\theta\sin\phi} + c_y \tag{20}$$

First-Order Expansion in R/Z_0

For a motor shaft, $R = 6$ mm and $Z_0 \approx 250$ mm, so $\varepsilon = R/Z_0 \approx 0.024 \ll 1$. Expanding (19) to first order:

$$\frac{1}{Z_0 - R\sin\theta\sin\phi} = \frac{1}{Z_0}\cdot\frac{1}{1-\varepsilon\sin\theta\sin\phi} \approx \frac{1}{Z_0}\left(1+\varepsilon\sin\theta\sin\phi\right) \tag{21}$$

Substituting into (19) and (20):

$$u(\phi) \approx \frac{f_xR}{Z_0}\cos\phi\left(1+\frac{R\sin\theta}{Z_0}\sin\phi\right) + c_x = \frac{f_xR}{Z_0}\cos\phi + \frac{f_xR^2\sin\theta}{Z_0^2}\cos\phi\sin\phi + c_x \tag{22}$$
$$v(\phi) \approx \frac{f_yR\cos\theta}{Z_0}\sin\phi\left(1+\frac{R\sin\theta}{Z_0}\sin\phi\right) + c_y = \frac{f_yR\cos\theta}{Z_0}\sin\phi + \frac{f_yR^2\sin\theta\cos\theta}{Z_0^2}\sin^2\phi + c_y \tag{23}$$

Ellipse Parameters

The second-order terms in (22)–(23) introduce an asymmetry in the ϕ-dependence that shifts the apparent ellipse center and inflates the semi-axes relative to the weak perspective case. Ignoring the $\cos\phi\sin\phi$ and $\sin^2\phi$ cross-terms (which contribute to center offset, not axis inflation), the dominant first-order terms give:

$$u \approx \frac{f_xR}{Z_0}\cos\phi + c_x, \qquad v \approx \frac{f_yR\cos\theta}{Z_0}\sin\phi + c_y \tag{24}$$

This is the weak perspective approximation: the image is an ellipse with semi-axes $a_0 = f_xR/Z_0$ (horizontal) and $b_0 = f_yR\cos\theta/Z_0$ (vertical), and the ratio $b_0/a_0 = (f_y/f_x)\cos\theta$.

For the full perspective case, retaining the $1/Z_c(\phi)$ term exactly and analyzing the extremal values of $u(\phi)$ and $v(\phi)$ yields the additional inflation factor $1/\cos\theta$ on both axes:

$$a = \frac{f_xR}{Z_0\cos^2\theta}, \qquad b = \frac{f_xR}{Z_0\cos\theta} \tag{25}$$

This inflation arises because the Z-coordinate varies along the tilted circle ($Z_c = Z_0 - R\sin\theta\sin\phi$), and points farther from the camera (larger Z_c) have smaller projected displacements, while points closer to the camera (smaller Z_c) have larger projected displacements. The net effect is an asymmetric magnification that stretches the apparent ellipse axes.

Scale Factor Derivation

From (25), we directly obtain $\cos\theta = b/a$. The pixel-to-metric scale Z_0/f_x is then:

$$\frac{Z_0}{f_x} = \frac{R}{b\cos\theta} = \frac{R}{b\cdot(b/a)} = \frac{Ra}{b^2} = \frac{Da}{2b^2} \tag{26}$$

which is the formula used in the main text.

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